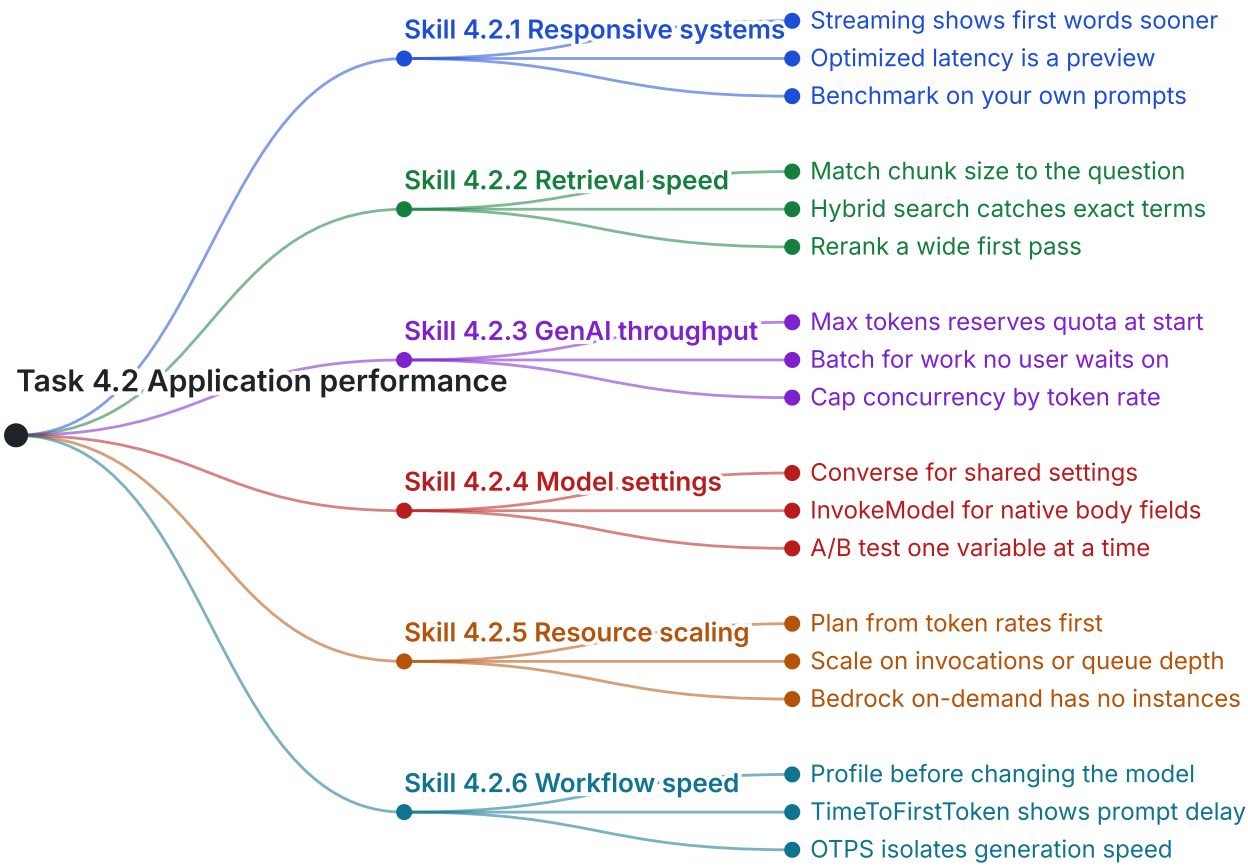
# Memory sheet: Task 4.2 Optimize application performance

Domain 4, weight 12 percent. Read this sheet, then answer the flash questions aloud.



## Deciding rules

| When the stem says | Think of | Not when |
|---|---|---|
| Chat, typing, or a live agent waiting | Interactive path with streaming | A batch job, because the user cannot wait hours |
| Overnight, backlog, or by morning, with data already in S3 | Batch inference on S3 files | A user waits for each reply |
| First words are slow, and text appears only at the end | Response streaming, measured by TimeToFirstToken | The program needs the complete object before it acts |
| Steady all day, or reserved | Provisioned or Reserved capacity | A spiky load, since idle hours still bill |
| Spikes, sales, or unpredictable traffic | On-demand with token budgets | A customized model that needs dedicated capacity |
| Exact codes, product numbers, or rare terms are missed | Hybrid search on a store with a filterable text field | A store with no filterable text field |
| The right passage is found but ranked low | Reranker over a wide first pass | The right passage never came back from search |
| 503 errors while quota use stays flat | Backoff with jitter and lower concurrency | A 400 error, which is an input problem |

## Etch lines

1. Match the symptom to its layer before you pick a tool.
2. Adding capacity does not shorten one unqueued call.
3. TimeToFirstToken is published only for streaming operations.
4. A larger max tokens value adds no context. It only reserves quota.
5. Retry 429, 503, and 529 with jitter. Never retry a 400.
6. A NAT path drops connections idle for 350 seconds.

## Flash questions

- Q: The first words from the assistant lag, yet throttle counts are zero. What is the first change to try? A: Turn on response streaming, then measure TimeToFirstToken.
- Q: Which metric covers the delay before the first token of a stream? A: TimeToFirstToken, which applies only to streaming operations.
- Q: Which metric separates generation speed from response length, and how is it built? A: OTPS. It divides output tokens by generation time, times 1000.
- Q: Does a bigger max tokens value give the model more input context? A: No. It raises the output ceiling and reserves more quota.
- Q: A passage sits in the index but ranks near the bottom of results. Which fix applies? A: A reranker over a wide first pass of candidates.
- Q: Exact product numbers are missed by semantic search. Which search mode helps, and what must the store support? A: Hybrid search, when the store has a filterable text field.

- Q: Requests fail with 503 and the quota graph is flat. Should the team ask for a quota increase? A: No. A 503 signals demand or capacity pressure, not a quota event.
- Q: A team needs one model-specific field that Converse cannot pass. Which API fits? A: InvokeModel, which sends the model-native JSON body.

Source: Skill sections 4.2.1 to 4.2.6, Network latency causes, and X versus Y confusable pairs. AWS docs and the study guide sections above, no practice-test text.